

Module 1: Activity 1 - Install and Configure the AWS CLI

Activity overview

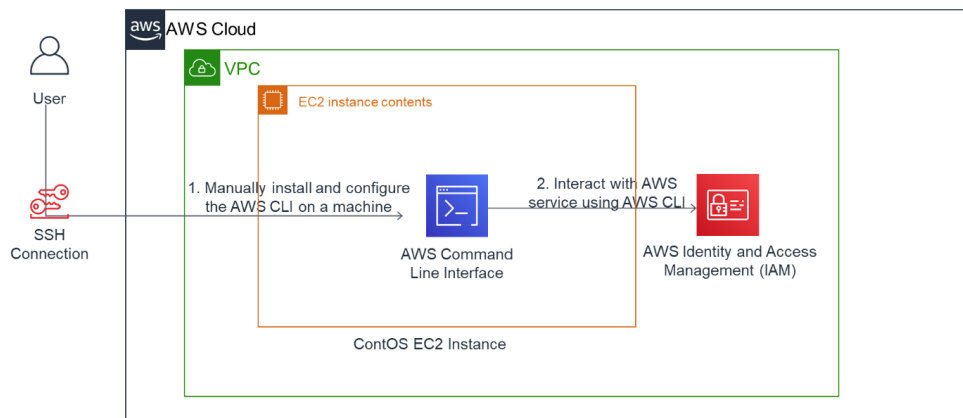
The *AWS Command Line Interface* (AWS CLI) is a command line tool that provides an interface for interacting with products and services from Amazon Web Services (AWS).

Often, people install the AWS CLI directly on their laptop machines. However, in this course—to ensure that all students have the same setup—you will practice using the AWS CLI from an Amazon Elastic Compute Cloud (Amazon EC2) instance.

While some instance types have the AWS CLI pre-installed on them (such as Amazon Linux instances), it is important to know how to install and configure this tool. Therefore, in this activity, you will practice installing the AWS CLI. A Red Hat Linux instance—that does not have the AWS CLI installed on it—is provided for you on Amazon EC2.

You will establish a Secure Shell (SSH) connection to the instance. Then, you will configure the installation with an access key that can connect to an AWS account. Finally, you will practice using the AWS CLI to interact with AWS Identity Access and Management (IAM).

This diagram summarizes the activities you will complete in this activity.



Activity objectives

After completing this activity, you will be able to:

- **Install** and configure the AWS CLI.
- **Connect** the AWS CLI to an AWS account.
- **Access** IAM by using the AWS CLI.

Business case relevance

Establishing AWS CLI Access for the Mom & Pop Café



After Sophie discussed with Matt, how to establish AWS CLI access to the Mom & Pop Café AWS account, she

After Sophie discussed with Matt, how to establish AWS CLI access to the Mom & Pop Café AWS account, she is confident that she can install, configure, and start making use of the AWS CLI.

During this activity, you will take on the role of Sophie. You will install the AWS CLI, configure it, and practice using the AWS CLI to query details about the IAM service.

Activity steps

Duration: This activity requires approximately **45 minutes** to complete.

Accessing the AWS Management Console

1. At the top of these instructions, choose to launch your lab.

A Start Lab panel opens displaying the lab status.

2. Wait until you see the message "**Lab status: ready**", then choose the **X** to close the Start Lab panel.

3. At the top of these instructions, choose

This will open the AWS Management Console in a new browser tab. The system will automatically log you in.

Tip: If a new browser tab does not open, there will typically be a banner or icon at the top of your browser indicating that your browser is preventing the site from opening pop-up windows. Choose the banner or icon and choose "Allow pop ups."

4. Arrange the AWS Management Console tab so that it displays along side these instructions. Ideally, you will be able to see both browser tabs at the same time, to make it easier to follow the lab steps.

Leave this browser tab open. You will return to it in Task 3.

Task 1: Connect to the Red Hat EC2 instance by using SSH

In this task, you will log in to an existing Amazon EC2 instance, which you will use to practice installing and using the AWS CLI.

Windows Users: Using SSH to Connect

 These instructions are for Windows users only.

If you are using macOS or Linux, skip to the next section.

5. Read through the three bullet points in this step before you start to complete the actions, because you will not be able to see these instructions when the Details panel is open.

- Choose the **Details** drop down menu above these instructions you are currently reading, and then choose **Show**. A Credentials window will open.
- Choose the **Download PPK** button and save the **labsuser.ppk** file. Typically your browser will save it to the Downloads directory.
- Then exit the Details panel by choosing the **X**.

6. Download needed software.

- You will use **PuTTY** to SSH to Amazon EC2 instances. If you do not have PuTTY installed on your computer, [download it here](#).

7. Open **putty.exe**

8. Configure PuTTY to not timeout:

- Choose **Connection**
- Set **Seconds between keepalives** to **30**

This allows you to keep the PuTTY session open for a longer period of time.

9. Configure your PuTTY session:

- Choose **Session**
- **Host Name (or IP address)**: Paste the *Public DNS or IPv4 address* of the Bastion Host instance that you noted earlier.
- Back in PuTTY, in the **Connection** list, expand **SSH**
- Choose **Auth** and expand **Credentials**
- Under **Private key file for authentication**: Choose **Browse**
- Browse to the *labsuser.ppk* file that you downloaded, select it, and choose **Open**
- Choose **Open** again

10. To trust and connect to the host, choose **Accept**.

11. When prompted **login as**, enter: `ec2-user`

This will connect you to the EC2 instance.

Task 2: Install the AWS CLI on Red Hat Linux

Now that you are connected to the instance, the next step is to install the AWS CLI.

For all steps that are in this section: Complete these steps in the terminal window where you have an active SSH connection to the Red Hat Linux instance running on Amazon EC2.

20. Verify that Python is installed by running the following command:

```
python3 --version
```

The command output shows that Python version 3.8.x is installed.

Tip: To install the AWS CLI, you must have Python 2 version, 2.6.5 or later, or Python 3 version 3.3 or later.

21. Run the following command to install the unzip utility.

```
sudo yum install -y unzip
```

22. Download and install AWS CLI :

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
unzip awscliv2.zip
sudo ./aws/install
```

23. Verify that AWS CLI is now working by running the following command:

```
aws help
```

The help command should display the help information for AWS CLI.

24. At the `:` prompt, type `q` to exit.

Task 3: Observe IAM configuration details in the AWS Management Console

25. Back in the **AWS Management Console** browser tab, choose **Services**, and then in the **Find a service...** search bar, type **IAM** to find the IAM service. Select it in the result set.

Note: The IAM page that displays will contain messages indicating that you do not have permission to observe some IAM service details. You can safely ignore these messages.

26. In the IAM service page, in the left navigation pane, choose **Users**, and then choose the **awsstudent** user, which will be a hyperlink.

27. Choose the arrow icon next to **lab_policy**, and then choose **{ } JSON** to see the policy document, which is formatted in JavaScript Object Notation (JSON).

Notice that this IAM policy grants the **awsstudent** user access to specific AWS services in this account.

28. Choose the **Security credentials** tab.

Notice that the access key was already created for the **awsstudent** user.

The **Access key ID** is displayed. There is no way to retrieve the *secret access key* that must be used with it, unless it was captured at the time that the key was created. Fortunately, the secret access key was captured when it was created, and it is available in the **Details** dropdown menu above. You will use these credentials in the next section of this activity.

Task 4: Configure the AWS CLI to connect to your AWS Account

29. Return to the active SSH session terminal window and run the configuration command for AWS CLI as follows:

```
aws configure
```

30. At the prompts, enter the following information:

- **AWS Access Key ID:** Choose the `Details` drop down menu above these instructions, and then choose `Show`. Copy the **AccessKey** value and paste it into the terminal window.
- **AWS Secret Access Key:** Copy and paste the **SecretKey** value from the same Credentials screen.
- **Default region name:** `us-east-1`
- **Default output format:** `json`

Task 5: Observe IAM configuration details by using the AWS CLI

31. In the terminal window, test the IAM configuration by running this command:

```
aws iam list-users
```

If the test is successful, you should see a JSON structured response that shows a list of all the IAM users in the account. The result should match what you saw in the AWS Management Console earlier in this activity.